ATAUR RAHMAN

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RESEARCH PROFILE

Mechanical engineering and advanced manufacturing researcher with an MSc in Advanced Manufacturing and research experience in sustainable polymer composites. Published in ACS Omega on the valorization of date palm agricultural residue in rotationally moulded LLDPE single- and multilayered composites and experienced in composite manufacturing, mechanical testing, thermal/materials characterisation, microscopy and engineering data analysis. Research interests include sustainable polymer composites, natural-fibre and waste-derived materials, circular manufacturing, advanced processing, interfacial behaviour, durability and engineering applications.

RESEARCH PUBLICATION

Rahman, A. et al. (2026). Date Palm Fiber Agricultural Residue Valorization in Rotationally Molded LLDPE Single- and Multilayered Composites for Engineering Applications. ACS Omega. DOI: 10.1021/acsomega.6c01931.

- Author contribution: methodology, investigation, formal analysis, visualization and data curation.
- Research focus: 5-20 wt% date palm fibre in LLDPE, single, double and triple-layer rotational moulding, rheology, SEM, tensile/flexural/impact testing, DSC, TGA, XRD, thermal conductivity and surface analysis.
- Key research direction identified by the study: fibre surface modification and long-term mechanical behaviour including fatigue, ageing and hygroscopic behaviour.

RESEARCH EXPERIENCE

Post-Graduate Research Associate | University of Portsmouth, UK | October2023 – June 2024

- Conducted research on sustainable plastic composites using Date Palm Fibre and polyethylene through rotational moulding.
- Prepared composite materials and operated rotational moulding and laboratory testing equipment using controlled experimental procedures.
- Performed tensile, flexural and Charpy impact testing and assessed thermal behaviour and material stability.

- Recorded, analysed and interpreted experimental data and contributed to technical research documentation.
- Worked on material development aimed at reducing reliance on petroleum-based polymers through agricultural-waste utilisation.

SELECTED RESEARCH PROJECT

Promoting Sustainability in the Rotational Moulding Plastic Sector by Utilising Natural Fibre Waste for the Manufacturing of Bio-Composites | University of Portsmouth

- Investigated Date Palm Fibre as a natural waste reinforcement for polyethylene-based composites.
- Developed an experimental understanding of fibre dispersion, mechanical behaviour and thermal performance in rotationally moulded composites.

Sustainability Approach in Dyeing Industries | BGMEA University of Fashion & Technology

- Evaluated dyeing-industry waste and investigated natural dyes and process-management strategies to improve sustainability.

EDUCATION

MSc Advanced Manufacturing | University of Portsmouth, UK | January 2023 – July 2024

- Key subjects: CAD, CAM, Quality Management, Simulation, Additive Manufacturing, Manufacturing Technology and Supply Chain.

BSc Apparel Manufacturing | BGMEA University of Fashion & Technology, Bangladesh | January 2016 – May 2020

- Key subjects: AutoCAD, Fibre-Yarn-Fabric, Textile Testing, Sewing Technology and Fashion Marketing.

TECHNICAL SKILLS

- Materials & manufacturing: rotational moulding, polymer composites, natural-fibre composites, sample preparation, mechanical testing, materials characterisation.
- Laboratory equipment: Zwick/Roell Z030E, manual and automatic Charpy impact testing machines, rotational moulding machine, ZM 300 grinding machine, SEM, DSC, TGA and DMA.
- Testing & analysis: tensile, flexural and impact testing, thermal analysis, morphology, crystallinity and engineering data analysis.
- Software: MATLAB, AutoCAD, SolidWorks, Enterprise Dynamics, K-kord, Microsoft Office.

INDUSTRY EXPERIENCE

Intern & Assistant Merchandiser | Aman Knitting's Ltd (Aman Graphics & Design Ltd), Dhaka | March 2020 – April 2022

- Worked with merchandising, production and quality teams to support manufacturing activities and customer requirements.
- Prepared costing and booking sheets, supported sample development and production-layout inspections.
- Communicated with internal and external stakeholders and monitored shipment progress.

Customer Assistant | Marks & Spencer, Chichester, UK | April 2025 – Present

- Work in a fast-paced environment, following procedures, safety requirements and quality standards while managing competing priorities.

ADDITIONAL

- Languages: Bengali (fluent), English (fluent), Hindi and Urdu (intermediate).
- Winner, Business Quiz Competition – BUFT Business Club (2017).
- Finalist, Book Reading Competition – British Council (2016).